

PROJECT:
REPAIR / REPLACE BASKETBALL FLOOR, 1201

PROJECT NUMBER:
MAHG251046



KEESLER AIR FORCE BASE
BILOXI, MISSISSIPPI

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INDEX

09 62 40	MODULAR ATHLETIC FLOORING
09 64 66	WOOD FLOORING – RAQUETBALL COURT
09 65 13	RESILIENT BASE AND ACCESSORIES

SECTION 09 62 40
MODULAR ATHLETIC FLOORING

PART 1 - GENERAL

1.1 DESCRIPTION

A. Scope

1. The complete installation of modular sports surfacing system including the interlocking suspended high-impact polypropylene copolymer tile of proprietary formulation, supportive acoustic underlayment and striping.
2. Provide striping and inserts, as required, for basketball, volleyball and pickleball as outlined in the SOW.

1.2 REFERENCES

A. ASTM (American Society for Testing & Materials)

1. ASTM G-21
2. ASTM D4060

B. DIN (Deutsches Institut für Normung)

1. DIN 18032-2

C. GMW (General Motors Worldwide Standards)

1. GMN3943 GMW14698

D. ISO (International Organization for Standardization)

1. ISO 9001:2015
2. ISO 14001:2015

1.3 SUBMITTALS

- A. Flooring system product data.
- B. Shop drawings.
- C. Product sample.
- D. Flooring Care and Maintenance Guide.
- E. Warranty.

1.4 QUALITY ASSURANCE

A. MATERIAL SUPPLIER:

1. Manufacturer must be ISO 9001:2015 & ISO 14001:2015 Certified to assure proper quality and environmental control.
2. Manufacturer shall have a minimum of 15 years' experience in specified product.
3. Product must have factory applied urethane coating.

B. INSTALLER:

1. The complete installation of the flooring system, as described in these specifications, shall be conducted by an experienced installer (Flooring Contractor), and the work shall be performed in accordance with current installation instructions of the manufacturer.
2. Contractor shall be liable for all matters related to installation for a period of one year after the floor has been substantially completed.
3. Successful bidder must submit a minimum of five (5) completed modular projects of similar magnitude and complexity within the last two (2) years.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Materials must be delivered in manufacturer's original, unopened and undamaged packaging with identification labels intact.
- B. Store material on a clean, dry, and flat surface, protected from exposure to harmful weather conditions or possible damage.
- C. Storage conditions shall be 55°F to 80°F (13°C to 27°C).

1.6 SITE CONDITIONS

- A. In order to prevent damage and not void the warranty, installation of modular materials shall not commence until all other finishes and overhead mechanical trades have completed their work in the modular floor areas.
- B. Permanent heat, light and ventilation shall be installed and operating during and after installation.
- C. Subfloors shall be clean, dry and free from dirt, dust, oil, grease, paint, old adhesive residue, or other foreign materials.
- D. Flooring installation shall not begin until the level of requirements of the concrete subfloors have been met.
- E. The installation area shall be closed to all traffic and activity for a period to be set by the flooring contractor.
- F. Product shall be conditioned at temperatures between 55°F to 80°F (13°C to 27°C) and shall be maintained for 24 hours prior to, during, and 24 hours after installation.
- G. Environmental Limitations
 - 1. Comply with the Manufacturer's requirements.
 - 2. Adhere to all SDS requirements for materials employed in the work.
 - 3. Protect all persons from exposure to hazardous materials at all times.
- H. After modular floors are installed and the game lines painted, the area is to be closed to allow curing time for the system, typically 3-5 days. No other trades or personnel are allowed on the floor until it has been accepted by the owner.

1.7 WARRANTY

- A. Provide a limited warranty of fifteen (15) years on the materials. A copy of the full warranty, with its Terms and Exclusions, shall be submitted.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Flooring system shall be:
 - 1. One piece solid-top design with optional shock absorbing sublayer up to 7mm.
 - 2. Metric-sized: 25cm x 25cm x 12.7mm (9.842" x 9.842" x ½").
 - 3. High-impact polypropylene copolymer suspended modules.
 - 4. 281 individual hexagonal cell support structure.
 - 5. Proprietary Maple or solid color in-mold foil transfer with a multi-layer factory applied, wear resistant polyurethane clear coat.
 - 6. The tile shall have a patented positive locking system.
- B. Standard Colors:
 - 1. Standard Maple Select™ Colors: Light Maple.
- C. Color Consistency: $\Delta E_{CMC} < 1.0$.
- D. Weight: 0.61 ± 0.01 lbs. (278 ± 5 grams)

- E. Packaging: Product is shipped in pre-assembled sheets (2x4 tiles per sheet, 6 sheets per box) or on pallets in pre-assembled 4x4 sheets of 16 tiles.
 - F. Product Test Results
 - 1. Force Reduction (DIN 18032-2): 0.20" (5mm underlayment) 25% - 30%.
(Shock Absorption) 0.35" (7mm underlayment) – 30% - 35%
 - 2. Ball Rebound (DIN 18032-2): 0.20" (5mm underlayment) – 100%
0.35" (7mm underlayment) – 100%
 - 3. Flatness: 0.0" +0.029" /-0.0" (0.0mm +0.74mm /-0.0mm)
 - 4. Mar Resistance (GMN3943 GMW14698): Rating 1 – No markings or insignificant damage when scratching, resistance to kick marks and burnishing.
 - 5. Abrasion Resistance (ASTM D4060): Less than 185 mg Mass Loss, >500 revolutions wear through, resistance to wear and use.
 - 6. Lateral Forgiveness™: +0.045" / -0.0" (+1.14mm / -0.0mm)
 - G. Load Bearing Capacity: 220 psi (1.52 MPa)
 - H. Dynamic Load - All systems must be able to show verification of passing a minimum 1,000,000 cycles in dynamic load testing with minimum 200 lbs. (91 kg) loading without deviation from flatness specification.
 - I. Underlayment – multi-purpose recycled rubber underlayment having a thickness of 0.20" (5mm) or 0.35" (7mm) and a Durometer of 60 ± 5 on the Shore A scale.
 - J. Retrofit – existing floor systems, synthetic floors must not exceed a maximum thickness of 7mm. Product must have a minimum durometer of 60 ± 5 on the Shore A scale. Flooring representative shall verify that the existing floor system meets these requirements by submitting core samples to the respective manufacturers for analysis and approval.
 - K. Sanitary Information
 - 1. Resistance to fungi (when tested in compliance with ASTM G-21 and MIL standard 810-D procedure 508.3). All basic organisms tested (ATCC #6205-11797) and were found to have zero growth.
 - 2. Resistance to the following:
 - a. Bacteria and mildew resistance
 - b. Gram-positive bacterial Staphylococcus Aureus
 - c. Gram-negative Klebsiella Pneumoniae
 - d. Pink-staining organism
 - e. STV Reticulum
 - f. Surface fungi growth prior to and following leaching.
 - L. Game Line Paint - aliphatic polyurethane as supplied by manufacture. Select from available colors.
- The striping corresponding to each sport shall be as follows: Basketball striping blue, Volleyball striping red, and Pickleball striping brown.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Inspect concrete slab for contamination, dryness and levelness. The slab shall be assessed for flatness. The required tolerance is a maximum deviation of $\pm 1/8$ inch within any 10-foot radius. Report any discrepancies to the general contractor.
- B. All surface imperfections shall be corrected by grinding high spots, filling low spots with an approved acrylic patching binder, and routing and filling cracks wider than $1/8$ inch with a flexible acrylic crack sealant

- C. Concrete slab shall be broom cleaned, mopped and dust free by the general contractor.
- D. Installer (Flooring Contractor) shall document all working conditions as specified in PART 1 – GENERAL prior to starting installation. Report any discrepancies to Buyer.

3.2 INSTALLATION

- A. Underlayment – Rubber underlayment shall be unrolled and allowed to relax. All butt joints shall be properly trimmed, fitted, and seamed together with an approved all-purpose tape.
- B. Floor shall be installed with a pre-approved layout.
- C. Minimum clearance at all vertical obstructions of 3/4 inch (19mm) is required.
- D. Floor surface shall be clean and dust free.
- E. Game Lines (Striping)
 - 1. Use only high-quality masking tape approved by the manufacturer.
 - 2. Lines shall be painted using the manufacturer's recommended aliphatic polyurethane paint.
 - 3. Provide game lines as indicated on drawings and as stated in the SOW.
 - 4. Room temperature shall be >55°F (13°C) and rising during paint installation.
- F. Wall Base - Install cove base adhered to walls with base cement.
- G. Remove all excess and waste materials from the area of work. Dispose of empty containers in accordance with federal and local statutes.

END OF SECTION

Section 09 64 66
WOOD FLOORING -
RACQUETBALL COURT

PART 1 – GENERAL

1.01 – Description

A. Scope:

1. Provide all labor, materials, and equipment necessary to complete the installation of racquetball court wood flooring. Striping for the court shall be red as stated in the SOW. All work shall be completed in a professional manner, conforming to industry standards, including but not limited to the following:

B. Quality Assurance:

Installer:

All work in conjunction with the racquetball court flooring shall be installed by one specialty contractor approved by the manufacturer.

1.02 – Submittals

- A. Material Samples
- B. Product data.
- C. Shop Drawings: Submit in accordance with plans and specifications.

1.03 – Product Delivery, Storage and Handling

- A. No materials shall be delivered to the jobsite until building is completely enclosed and weathertight.
- B. Unloading and stocking materials into the court area to be the responsibility of the general contractor or owner unless otherwise arranged.
- C. It is critical to the acclimation of the flooring and panels that proper humidity and temperature be maintained in the court area. Therefore, the permanent HVAC shall be in operation and maintaining court area temperature between 50-80 degrees and 40%-60% humidity prior to and continuously after installation.
- D. Store materials properly stacked in a clean, dry, storage area.
- E. Material is not properly protected and stored, and which is defaced or damaged shall be rejected.

1.04 – Warranty

- A. Installation workmanship and maple floor system is warranted for a period of one year from date of installation. Warranty applies only when materials are

installed and environmental conditions are following manufacturers' specifications.

PART 2 – PRODUCTS

2.01 – Materials

A. Floor:

1. Existing concrete sub-floor shall be broom and mopped clean. Verify is any leveling required to meet this tolerance. Provide leveling if required.
2. Flooring shall be MFMA hard maple standard T&G strip flooring, 25/32" x 2-1/4".
3. Grades 1 or better.
4. Wood flooring may be Woodlife F treated when specified.
5. Vapor retarder shall be 6-mil polyethylene.
6. Sleepers shall be HPS laminated 2" x 3" x 8'-0" (38mm x 64mm x 2.4m), each with eight pads attached. Sleepers are to be installed end to end in rows 9" O.C. in a staggered configuration at right angles to the intended direction of the finished flooring with end joints overlapping 24".
7. Flooring fasteners shall be 2" cleats or 2" staples.
8. Gap of approx. 1/2" between floor and sidewall panels to be left for air circulation and floor expansion.
9. Floor to be sanded with a drum sander using coarse, medium and fine sandpaper. Vacuum or tack floor before first coat of finish.
10. Apply urethane finish per manufacturers specifications.
11. Paint game lines per regulations.

PART 3 – EXECUTION

3.01 – Inspection

- A. Inspect court area to ensure conditions as set forth in these specifications have been adhered to.
- B. Inspect concrete sub-floors for proper tolerance and dryness. Discrepancies should be reported to the Buyer.

3.02 – Installation

- A. Install court flooring per construction documents and manufacturer's written instructions and shop drawings.
- B. Clean up all debris caused by work in this section and remove the onsite dumpster provided by general contractor or owner.

END OF SECTION

SECTION 09 65 13

RESILIENT BASE AND ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Resilient base (RB) adhered to interior walls and partitions.

1.2 RELATED REQUIREMENTS

1.3 APPLICABLE PUBLICATIONS

A. Comply with references to extent specified in this section.

B. ASTM International (ASTM):

F1861-16..... Resilient Wall Base.

1.4 SUBMITTALS

A. Submittals in accordance with the Statement of Work.

B. Manufacturer's Literature and Data:

1. Description of each product.
2. Adhesives and primers indicating manufacturer's recommendation for each application.
3. Installation instructions.

C. Samples:

1. Resilient Base: 150 mm (6 inches) long, each type and color.

D. Sustainable Construction Submittals:

1. Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.
2. Low Pollutant-Emitting Materials:
 - a. Show volatile organic compound types and quantities.

E. Operation and Maintenance Data:

1. Care instructions for each exposed finish product.

1.5 DELIVERY

- A. Deliver products in manufacturer's original sealed packaging.
- B. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.
- C. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.6 STORAGE AND HANDLING

- A. Store products indoors in dry, weather tight facility.
- B. Protect products from damage when handling and during construction operations.

1.7 FIELD CONDITIONS

- A. Environment:
 - 1. Product Temperature: Minimum 21 degrees C (70 degrees F) for minimum 48 hours before installation.
 - 2. Work Area Ambient Temperature Range: 21 to 27 degrees C (70 to 80 degrees F) continuously, beginning 48 hours before installation.
 - 3. Install products when building is permanently enclosed and when wet construction is completed, dried, and cured.

1.8 WARRANTY

- A. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

PART 2 - PRODUCTS

2.1 PRODUCTS

- A. Provide each product from one manufacturer and from one production run.
- B. Sustainable Construction Requirement:
 - 1. Low Pollutant-Emitting Materials: Comply with VOC limits specified, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:
 - a. Resilient Base.

2.2 RESILIENT BASE

- A. Resilient Base: 6 inches high.
 - 1. Type: Rubber or vinyl; use one type throughout.
 - 2. ASTM F1861, Type TP thermoplastic rubber or Type TV thermoplastic vinyl, Group 2 - layered.
- B. Applications:
 - 1. Carpet Flooring Locations: Style A – Cove.

2.3 ADHESIVES

- A. Adhesives: Low pollutant-emitting, water-based type recommended by adhered product manufacturer for each application.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Examine and verify substrate suitability for product installation.
- B. Protect existing construction and completed work from damage.
- C. Remove existing base to permit new installation.
 - 1. Dispose of removed materials.
- D. Clean substrates. Remove contaminants capable of affecting subsequently installed products' performance.

3.2 INSTALLATION GENERAL

- A. Install products according to manufacturer's instructions.
 - 1. When instructions deviate from specifications, submit proposed resolution for Contracting Officer consideration.

3.3 RESILIENT BASE INSTALLATION

- A. Applications:
 - 1. Install resilient base in rooms scheduled on Drawings.
 - 2. Install resilient base on casework, and other curb supported fixed equipment.
 - 3. Extend resilient base into closets, alcoves, and cabinet knee spaces, and around columns within scheduled room.
- B. Lay out resilient base with minimum number of joints.
 - 1. Length: 24 inches minimum, each piece.
 - 2. Locate joints 6 inches minimum from corners and intersection of adjacent materials.
- C. Installation:
 - 1. Apply adhesive uniformly for full contact between resilient base and substrate.
 - 2. Set resilient base with hairline butted joints aligned along top edge.
- D. Field form corners and end stops.
 - 1. V-groove back of outside corner.
 - 2. V-groove face of inside corner and notch cove for miter joint.
- E. Roll resilient base ensuring complete adhesion.

3.4 CLEANING

- A. Remove excess adhesive before adhesive sets.

B. Clean exposed resilient base. Remove contaminants and stains.

1. Clean with mild detergent. Leave surfaces free of detergent residue.

3.5 PROTECTION

A. Protect products from construction traffic and operations.

1. Maintain protection until directed by Contracting Officer's Representative.

B. Replace damaged products and reclean.

1. Damaged Products include cut, gouged, scraped, torn, and un-bonded products.

END OF SECTION